AIR UNIVERSITY, KAMRA
Department of Computer Science
Computer Networks (CN) — Project Proposal

PROJECT PROPOSAL

University Campus Network Design & Implementation

Cisco Packet Tracer Simulation

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1. Executive Summary

This proposal outlines the design and implementation plan for a fully functional University Campus Network using Cisco Packet Tracer. The project aims to simulate a real-world hierarchical network infrastructure for Air University, Kamra, interconnecting multiple zones including computer laboratories, lecture rooms, administrative offices, an examination office, and a centralized server room.

The network will demonstrate practical application of core Computer Networks concepts including static IP addressing, inter-zone routing, fault-tolerant ring topology, DNS resolution, HTTP web hosting, email services, Access Control Lists (ACLs), and wireless connectivity. The simulation will serve as a complete, testable model that mirrors enterprise-grade campus network architecture.

2. Project Objectives

The primary goals of this project are as follows:

- Design a scalable, hierarchical campus network for a multi-zone university building
- Implement static IP addressing with clear subnet isolation per zone
- Deploy DNS and HTTP servers to simulate real-world web hosting and domain resolution
- Establish secure inter-zone communication using Access Control Lists (ACLs)
- Achieve fault-tolerant routing using a ring topology with floating static routes
- Configure email services (SMTP/POP3) for internal university communication
- Deploy wireless access points for mobile device connectivity
- Validate all configurations through systematic connectivity and failover testing

3. Scope of Work

The project encompasses the complete lifecycle of network design, configuration, and verification within the Cisco Packet Tracer simulation environment. The scope is defined across the following functional areas:

3.1 Network Zones

The campus network will span the following distinct zones, each isolated in its own subnet:

Zone	Subnet	Gateway
Computer Lab 1	192.168.10.0/24	192.168.10.1
Computer Lab 2	192.168.20.0/24	192.168.20.1
Computer Lab 3	192.168.30.0/24	192.168.30.1
Computer Lab 4	192.168.40.0/24	192.168.40.1
Faculty Office	192.168.50.0/24	192.168.50.1
Upper Classrooms	192.168.60.0/24	192.168.60.1
Examination Office	192.168.70.0/24	192.168.70.1
Lower Classrooms	192.168.80.0/24	192.168.80.1
HOD Office	192.168.90.0/24	192.168.90.1
Server Room	192.168.100.0/24	192.168.100.1

3.2 Network Hardware (Simulated)

- 1x Central Router — Cisco 2911 (backbone hub)
- 6x Zone Routers — Cisco 2911 with HWIC-2T serial modules
- 10x Access Layer Switches — Cisco 2960-24TT
- Multiple Server-PT devices (DNS, HTTP, Email)
- End-user PCs, Laptops, and Smartphones per zone
- Wireless Access Points for Wi-Fi coverage

4. Technical Approach

4.1 Topology Design

The network will implement a hybrid Star-Ring topology. A central Cisco 2911 router forms the backbone and connects via serial WAN links to six zone routers. The six zone routers are additionally interconnected in a ring formation using serial DCE cables, providing redundant paths in the event of a central router failure.

4.2 IP Addressing Strategy

All devices will be assigned static IP addresses for reliability and ease of troubleshooting. LAN segments will use the 192.168.X.0/24 address space. Point-to-point WAN links between routers will use /30 subnets from the 10.0.X.0 and 10.1.X.0 address blocks, providing exactly two usable host addresses per link.

4.3 Routing Protocol

Static routing with floating static routes will be employed throughout the network. Each zone router will maintain three routing tiers using Administrative Distance (AD) values: primary paths via the central router (AD 1), first backup paths via the nearest ring neighbor (AD 10), and second backup paths via the opposite ring direction (AD 20). This ensures seamless failover without dynamic protocol overhead.

4.4 Server Room Services

The centralized server room will host the following services:

- DNS Server — A-record resolution for all hosted domains
- HTTP Servers — Simulated websites including www.aack.au.edu.pk, www.google.com, www.github.com, www.yahoo.com, www.chatgpt.com, and www.classroom.google.com
- Email Server — SMTP (port 25) and POP3 (port 110) for internal university communication under the domain aack.au.edu.pk

4.5 Access Control Lists (ACLs)

Extended Named ACLs will be configured to enforce a realistic university access hierarchy. Labs and classrooms will be permitted access to the Server Room and Faculty Office, but will be blocked from directly contacting the HOD Office and Examination Office. The HOD Office will have outbound access to all zones, but inbound replies from labs will be restricted — simulating a one-way announcement model. Exam Office access will be tightly controlled, permitting only HOD Office and Server Room communication.

4.6 Wireless Connectivity

Wireless access points will be integrated into selected zones, sharing the same subnet and default gateway as wired devices. This will allow laptops and smartphones to participate in all network functions, including inter-zone communication and server access, without any additional routing configuration.

5. Project Deliverables

- Fully operational Cisco Packet Tracer simulation file (.pkt)
- Complete static IP and routing configuration across all 7 routers and 10 switches
- Functioning DNS, HTTP, and Email services in the server room
- Verified ACL policies enforcing the university access hierarchy

- Fault-tolerant ring backup routing with documented failover testing
- Wireless connectivity demonstrated via laptops and smartphones
- Comprehensive project report documenting design, configuration, and test results

6. Proposed Timeline

Phase	Task	Duration
1	Network topology design and IP addressing scheme	Week 1
2	Router and switch configuration (static IPs, interfaces)	Week 2
3	Static routing and floating backup route setup	Week 3
4	Server room configuration (DNS, HTTP, Email)	Week 4
5	ACL design and implementation	Week 5
6	Wireless access point deployment	Week 6
7	Full testing, verification, and troubleshooting	Week 7
8	Report writing and submission	Week 8

7. Tools & Technologies

- Cisco Packet Tracer — Primary simulation environment
- Cisco 2911 Router — Core and zone routing
- Cisco 2960-24TT Switch — Access layer switching
- HWIC-2T Serial Module — WAN inter-router links
- Cisco Wireless Access Point — Wi-Fi simulation
- Server-PT — DNS, HTTP, and Email service simulation

8. Key Networking Concepts Applied

- Subnetting and CIDR notation (/24 LAN, /30 WAN point-to-point links)
- Hierarchical three-layer network design (Core, Distribution, Access)
- Static routing with Administrative Distance for fault tolerance
- Ring topology for redundant backup path routing

- Extended Named Access Control Lists for traffic filtering
- DNS A-record configuration and domain name resolution
- HTTP web server hosting within a private network
- SMTP and POP3 email service configuration
- Wireless LAN integration with wired network infrastructure

9. Conclusion

This project proposal presents a comprehensive plan to design, configure, and simulate a fully functional university campus network using Cisco Packet Tracer. The proposed network architecture will cover all major functional areas of a real university building, applying industry-standard networking concepts studied throughout the Computer Networks course.

The successful completion of this project will produce a fully verified, fault-tolerant network simulation that demonstrates practical competence in network design, IP addressing, routing, security, and service configuration. The project is well within the scope of the available simulation tools and the concepts covered in the course curriculum.

I look forward to your approval to proceed with this project and am committed to delivering a high-quality simulation and report within the proposed timeline.

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